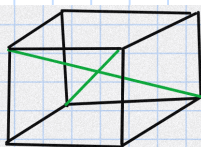
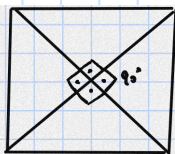
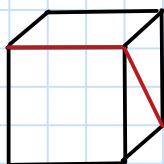


1)



a) Calcolare la diagonale

b) Se sono su un vertice del cubo, qual è il cammino minimo che posso fare per arrivare al vertice opposto muovendami sulla superficie?



passando per la diagonale di un lato e su un lato avrà un cammino minimo di 2, mentre percorrendo solo lati avrà sempre almeno 3

2) Dimostrare che, dato un triangolo arbitrario, prese le sue tre mediane, esiste sempre un triangolo i cui lati sono le tre mediane (costruisci un triangolo estraendo le tre mediane).

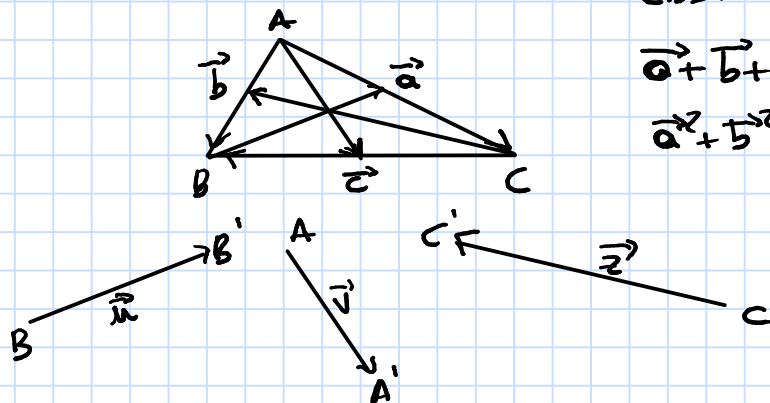
$$\vec{CB} = \vec{AB} - \vec{AC}$$

$$\vec{a} + \vec{b} + \vec{c} = 0$$

$$\vec{a} + \vec{b} = -\vec{c}$$

tutti i triangoli

triangoli rettangoli



$$B' = \frac{1}{2}(A+C)$$

$$A' = \frac{1}{2}(B+C)$$

$$C' = \frac{1}{2}(A+B)$$

$$\vec{u} = \frac{1}{2}(A+C) - B$$

$$\vec{v} = \frac{1}{2}(B+C) - A$$

$$\vec{z} = \frac{1}{2}(A+B) - C$$

la somma di questi tre segmenti orientati deve essere zero in quel caso avrà dimostrato che con le tre mediane del triangolo arbitrario posso costruire sempre un altro triangolo:

$$\vec{u} + \vec{v} + \vec{z} = 0$$

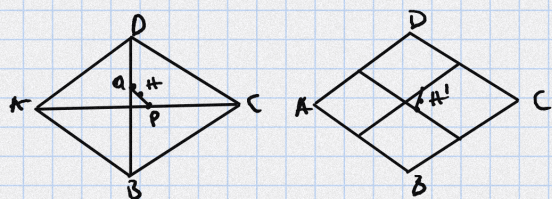
$$\frac{1}{2}(A+C) - B + \frac{1}{2}(B+C) - A + \frac{1}{2}(A+B) - C = 0$$

$$\frac{1}{2}A + \frac{1}{2}C - B + \frac{1}{2}B + \frac{1}{2}C - A + \frac{1}{2}A + \frac{1}{2}B - C = 0$$

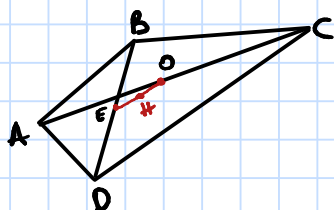
$$A - A + B - B + C - C = 0$$

verificato

3) Preso un quadrilatero, dimostra che il punto medio del segmento che unisce i punti medi delle due diagonali coincide con il punto medio del segmento che unisce i punti medi dei segmenti che uniscono i punti medi dei lati del quadrilatero.



$H = H'$
chi è H ?



$$H = \frac{1}{2}(E + O)$$

$$H' = \frac{1}{2}(R + S)$$

$$E = \frac{1}{2}(B + D) \quad O = \frac{1}{2}(A + C)$$

$$S = \frac{1}{2}(H + K) \quad R = \frac{1}{2}(F + G)$$

$$H = \frac{1}{2}(A + B) \quad K = \frac{1}{2}(D + C) \quad F = \frac{1}{2}(A + D) \quad G = \frac{1}{2}(B + C)$$

$$\frac{1}{2}(E + O) = \frac{1}{2}(R + S)$$

$$\frac{1}{2}\left(\frac{1}{2}(B + D) + \frac{1}{2}(A + C)\right) = \frac{1}{2}\left(\frac{1}{2}(F + G) + \frac{1}{2}(H + K)\right)$$

$$\frac{1}{2}\left(\frac{1}{2}(B + D) + \frac{1}{2}(A + C)\right) = \frac{1}{2}\left(\frac{1}{2}\left[\frac{1}{2}(A + D) + \frac{1}{2}(B + C)\right] + \frac{1}{2}\left[\frac{1}{2}(A + B) + \frac{1}{2}(D + C)\right]\right)$$

$$\frac{1}{2}\left[\frac{1}{2}B + \frac{1}{2}D + \frac{1}{2}A + \frac{1}{2}C\right] = \frac{1}{2}\left\{\frac{1}{2}\left[\left(\frac{1}{2}A + \frac{1}{2}D + \frac{1}{2}B + \frac{1}{2}C\right) + \left(\frac{1}{2}A + \frac{1}{2}B + \frac{1}{2}D + \frac{1}{2}C\right)\right]\right\}$$

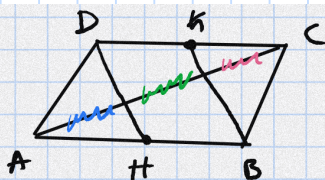
$$\frac{1}{4}B + \frac{1}{4}D + \frac{1}{4}A + \frac{1}{4}C = \frac{1}{2}\left\{\frac{1}{4}A + \frac{1}{4}D + \frac{1}{4}B + \frac{1}{4}C + \frac{1}{4}A + \frac{1}{4}B + \frac{1}{4}D + \frac{1}{4}C\right\}$$

$$\frac{1}{4}B + \frac{1}{4}D + \frac{1}{4}A + \frac{1}{4}C = \frac{1}{2}\left\{\frac{1}{2}A + \frac{1}{2}D + \frac{1}{2}B + \frac{1}{2}C\right\}$$

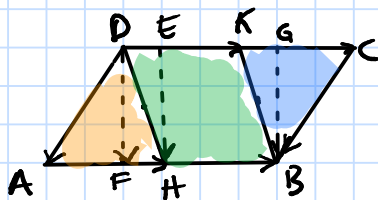
$$\frac{1}{4}B + \frac{1}{4}D + \frac{1}{4}A + \frac{1}{4}C = \frac{1}{4}B + \frac{1}{4}D + \frac{1}{4}A + \frac{1}{4}C$$

verificato

4)



Dimostrare che i segmenti $\overline{DH}$ e $\overline{KB}$ dividono il parallelogramma in tre parti uguali



$$\Rightarrow \vec{AH} = \vec{DK} = \vec{KC}$$

$$\vec{AH} = \frac{1}{2}(\vec{A} + \vec{B}) - \vec{A}$$

$$\vec{DK} = \frac{1}{2}(\vec{D} + \vec{C}) - \vec{D}$$

$$\vec{KC} = \vec{C} - \frac{1}{2}(\vec{D} + \vec{C})$$

$$\frac{1}{2}(\vec{A} + \vec{B}) - \vec{A} = \frac{1}{2}(\vec{D} + \vec{C}) - \vec{D} = \vec{C} - \frac{1}{2}(\vec{D} + \vec{C})$$

$$\frac{1}{2}\vec{A} - \vec{A} + \frac{1}{2}\vec{B} = \frac{1}{2}\vec{D} - \vec{D} + \frac{1}{2}\vec{C} = \vec{C} - \frac{1}{2}\vec{D} - \frac{1}{2}\vec{C}$$

$$\frac{1}{2}\vec{A} - \vec{A} + \frac{1}{2}\vec{B} = \frac{1}{2}\vec{D} - \vec{D} + \frac{1}{2}\vec{C}$$

$$-\frac{1}{2}\vec{A} + \frac{1}{2}\vec{B} = -\frac{1}{2}\vec{D} + \frac{1}{2}\vec{C}$$

allora

$$-\frac{1}{2}\vec{A} + \frac{1}{2}\vec{B} = \vec{C} - \frac{1}{2}\vec{D} - \frac{1}{2}\vec{C}$$

$$-\frac{1}{2}\vec{A} + \frac{1}{2}\vec{B} = -\frac{1}{2}\vec{D} + \frac{1}{2}\vec{C}$$

verificato che sono divisi in tre parti uguali i lati

Ora calcolo l'area dei tre pezzi e verifico che hanno tre parti uguali:

$$\hat{A} \downarrow \text{DH} \\ A_1 = \frac{\vec{AH} \cdot \vec{DF}}{2}$$

$$\hat{K} \downarrow \text{BC} \\ A_2 = \frac{\vec{KC} \cdot \vec{GB}}{2}$$

$$\hat{K} \downarrow \text{BDH} \\ A_3 = \frac{\vec{HE} \cdot \vec{DK}}{2}$$

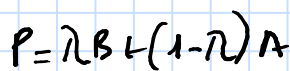
$$\vec{DF} = \vec{GB} = \vec{HE}$$

$$\begin{aligned} \vec{DF} &= \vec{F} - \vec{D} \\ \vec{GB} &= \vec{B} - \vec{G} \\ \vec{HE} &= \vec{E} - \vec{H} \end{aligned}$$

$$A_1 = \frac{\left[\frac{1}{2}(\vec{A} + \vec{B}) - \vec{A} \right] \left[\vec{F} - \vec{D} \right]}{2} = \frac{\left(-\frac{1}{2}\vec{A} + \frac{1}{2}\vec{B} \right) (\vec{F} - \vec{D})}{2} = \frac{-\frac{1}{2}\vec{AF} + \frac{1}{2}\vec{AD} + \frac{1}{2}\vec{BF} - \frac{1}{2}\vec{BD}}{2} = \frac{1}{4}(-\vec{AF} + \vec{AD} + \vec{BF} - \vec{BD})$$

$$A_2 = \left[-\frac{1}{2}\vec{D} + \frac{1}{2}\vec{C} \right] \left[\vec{B} - \vec{G} \right] = -\frac{1}{2}\vec{BD} - \frac{1}{2}\vec{CG} = -\frac{1}{2}(\vec{BD} + \vec{CG})$$

$$A_3 = \frac{\left[-\frac{1}{2}\vec{D} + \frac{1}{2}\vec{C} \right] \left[\vec{E} - \vec{H} \right]}{2} = \frac{-\frac{1}{2}\vec{DE} + \frac{1}{2}\vec{DH} + \frac{1}{2}\vec{CE} - \frac{1}{2}\vec{CH}}{2} = \frac{1}{4}(-\vec{DE} + \vec{DH} + \vec{CE} - \vec{CH})$$

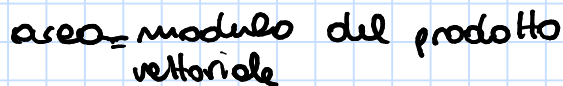


$$Y = \lambda_H (1 - \lambda) D$$

$$\vec{AY} = Y - A = \lambda H + (1 - \lambda)D - A = \lambda H + D - \lambda D - A$$

$$\overline{XC} = C - X = C - [\lambda B + (1-\lambda)K] = C - \lambda B - K + \lambda K$$

$$\lambda_H - \lambda_D - \lambda_B + \lambda_K = K - Y + A - D$$



Area = metà del prodotto vettoriale $\frac{|\vec{e} \times \vec{p}|}{2}$

$$\vec{b} = H \hat{j}$$

$$\vec{a} = D\hat{j} + D\hat{k}$$

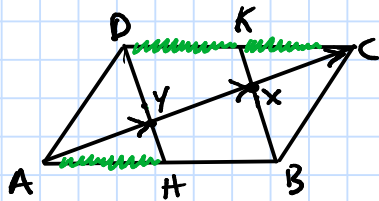
$$O_{ij} (-1)^{i+j}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 0 \\ 0 & 4 & 0 \end{vmatrix} = -\frac{40\hat{j}}{2}$$

$$\vec{c} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ H & 0 & H \\ K & 0 & K \end{vmatrix} = -\hat{j}(HK - KH)$$

$$\vec{e} \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \beta & \beta & 0 \\ \kappa & 0 & \kappa \end{vmatrix} = \frac{\hat{i}\beta\kappa - \hat{j}\beta\kappa - \hat{k}\beta\kappa}{2}$$

$$\vec{p} = \hbar \hat{k} + K \hat{R}$$



$$\overline{DK} = \frac{1}{2}(\overline{D+C}) - \overline{D}$$

$$\overline{KB} = \overline{K} - \overline{B} = \frac{1}{2}(\overline{D+C}) - \overline{B} = \overline{DH} = \overline{D-H} = \overline{D} - \frac{1}{2}(\overline{A+B})$$

$$H_p.: \overrightarrow{XC} = \overrightarrow{XY} = \overrightarrow{YA}$$

$$X = \lambda B + (1-\lambda)K$$

$$Y = \lambda H + (1-\lambda)D$$

$$\overrightarrow{XY} = X - Y = \lambda B + (1-\lambda)K - [\lambda H + (1-\lambda)D]$$

$$\overrightarrow{XY} = \lambda B + K - \lambda K - \lambda H - D + \lambda D = (B-K-H+D)\lambda + K-D$$

$$(B-K-H+D)\lambda + K-D = C - [\lambda B + (1-\lambda)K] = \overrightarrow{CX}$$

$$(B-K-H+D)\lambda + K-D = C - \lambda B - K + \lambda K$$

$$(B-K-H+D)\lambda + K-D = (K-B)\lambda + C-K$$

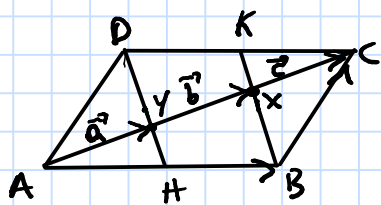
$$(B-K-H+D-K+B)\lambda = C-K-K+D$$

$$(2B-2K-H+D)\lambda = C+D-2K$$

$$\lambda = \frac{C+D-2K}{(2B-2K-H+D)}$$

se ottenuto lo stesso risultato con l'altra uguaglianza allora ho verificato

$$(B-K-H+D)\lambda + K-D = \lambda H + (1-\lambda)D - A$$



$$(\vec{a} \parallel \vec{b} \parallel \vec{c}) \parallel \vec{AC} \quad \text{con} \quad \vec{AC} = \vec{AB} + \vec{BC}$$

$$\vec{a} = \lambda \vec{b}$$

$$\vec{b} = \lambda \vec{c}$$

$$\vec{c} = \lambda \vec{a}$$

$$\text{H.p.: } |\vec{a}| = |\vec{b}| = |\vec{c}|$$

$$\text{Tesi: } \vec{a}, \vec{b} \text{ e } \vec{c} = \lambda \vec{AC} \quad \text{con} \quad \lambda = \frac{1}{3}$$

$$\vec{AC} = \vec{AB} + \vec{BC} = (B-A) + (B+C) = B-A+B+C = 2B-A+C$$

$$\vec{a} = H-A = \lambda H + (1-\lambda)D - A \quad \text{con} \quad H = \lambda H + (1-\lambda)D$$

$$\vec{b} = X-H = (B-K-H+D)\lambda + K-D \quad \text{con} \quad X = \lambda B + (1-\lambda)K$$

$$\vec{c} = C-X = (K-B)\lambda + C-K$$

Alora:

$$\lambda H + (1-\lambda)D - A = \lambda \vec{AC}$$

$$(B-K-H+D)\lambda + K-D = \lambda \vec{AC}$$

$$(K-B)\lambda + C-K = \lambda \vec{AC}$$

$$\lambda H + D - \lambda D - A = \lambda(2B-A+C)$$

$$(B-K-H+D)\lambda + K-D = \lambda(2B-A+C)$$

$$(K-B)\lambda + C-K = \lambda(2B-A+C)$$

$$\lambda(H-D-2B+A-C) = A-D$$

$$(B-K-H+D-2B+A-C)\lambda = D-K$$

$$(K-B-2B+A-C)\lambda = K-C$$

$$\lambda = \frac{A-D}{H-D-2B+A-C}$$

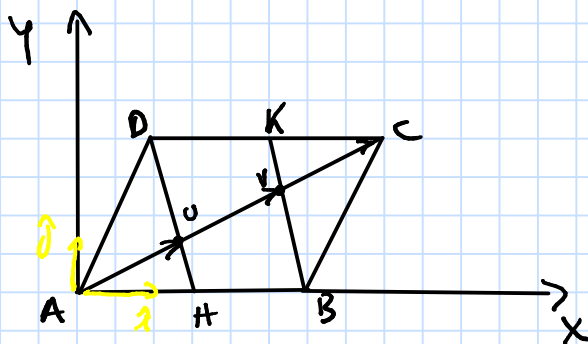
$$\lambda = \frac{D-K}{-B-K-H+D+A-C}$$

$$\lambda = \frac{K-C}{K-3B+A-C}$$

$$\lambda = \frac{\vec{DA}}{\vec{DH} + \vec{CA} - 2\vec{B}}$$

$$\lambda = \frac{\vec{KD}}{\vec{HD} + \vec{BA} + \vec{K}}$$

$$\lambda = \frac{\vec{CK}}{\vec{CK} + A - 3B}$$



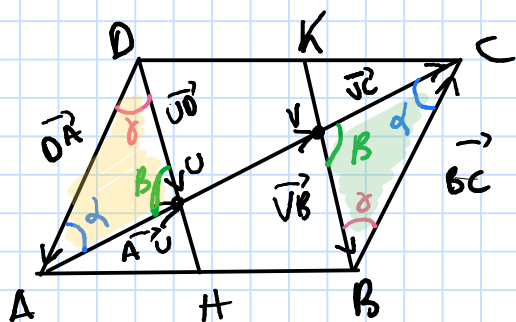
Ipotesiamo $C = 8x + 4y = C_x x + C_y y$

$$K = 5x + 4y = K_x x + K_y y$$

$$B = 6x = B_x x$$

$$V = \lambda(B_x x) + (1-\lambda)(K_x x + K_y y) = \lambda B_x x + K_x x - K_x x \lambda + K_y y - \lambda K_y y =$$

=



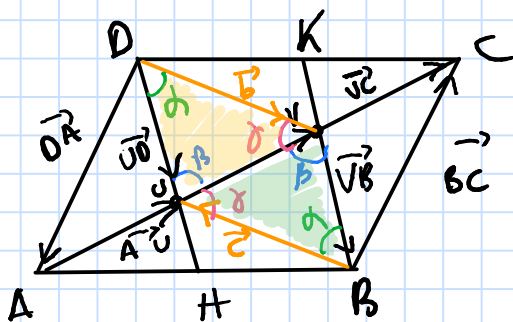
$$\frac{\vec{VB}}{\sin \alpha} = \frac{\vec{BC}}{\sin \beta} = \frac{\vec{VC}}{\sin \gamma}$$

$$\text{H.p.: } \vec{AU} = \vec{VC}$$

$$\frac{\vec{UD}}{\sin \alpha} = \frac{\vec{DA}}{\sin \beta} = \frac{\vec{AV}}{\sin \gamma}$$

$$\frac{\vec{AU}}{\sin \gamma} = \frac{\vec{VC}}{\sin \gamma} \rightarrow \vec{AU} = \vec{VC}$$

grazie agli angoli simili si può dimostrare che questi due vettori hanno stesso modulo



$$\frac{\vec{UV}}{\sin \alpha} = \frac{\vec{DU}}{\sin \beta} = \frac{\vec{UD}}{\sin \gamma}$$

$$\rightarrow \vec{UD} = \vec{VB}$$

$$\frac{\vec{UV}}{\sin \alpha} = \frac{\vec{BV}}{\sin \beta} = \frac{\vec{VB}}{\sin \gamma}$$

Se $\vec{UD}$ è la mediana di $\triangle ADU$ e $\vec{VB}$ è la mediana di $\triangle UBC$ allora anche $\vec{UV} = \vec{AU} = \vec{VC}$

Se $\vec{UD}$ è la mediana allora U è il punto medio di $\vec{AD}$

$$U = \frac{1}{2}(A+D)$$

$$\vec{UD} = \vec{OA} + \vec{AU} = \vec{OA} + \frac{1}{2}\vec{AD}$$

$$\vec{VB} = \vec{BC} - \vec{VC} = \vec{BC} - \frac{1}{2}\vec{UC}$$

$$\vec{UD} = \vec{VB} \text{ già verificato}$$

$$V = \frac{1}{2}(U+C)$$

$$\vec{DA} + \frac{1}{2}\vec{AV} = \vec{BC} - \frac{1}{2}\vec{UC}$$

con $\vec{DA} = \vec{BC}$ perché è un parallelogramma

$$\vec{DA} + \frac{1}{2}\vec{AV} = \vec{DA} - \frac{1}{2}\vec{UC}$$

$$\vec{AV} = \vec{AV} + \vec{UV} \quad \text{e} \quad \vec{UC} = \vec{UV} + \vec{VC}$$

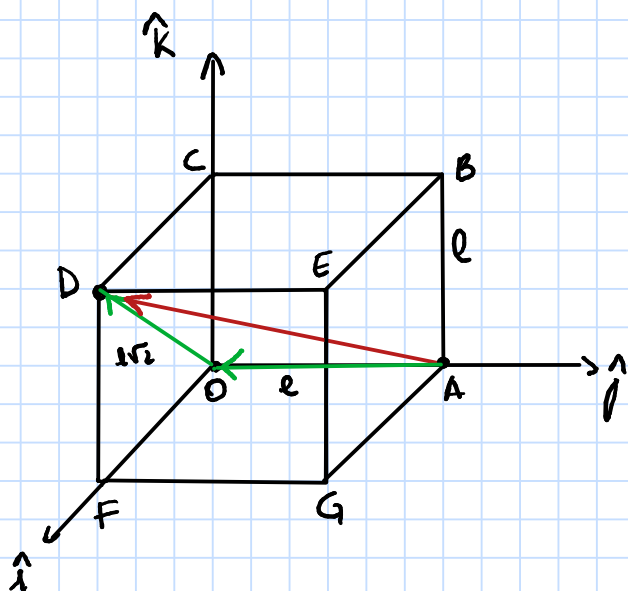
$$\vec{DA} + \frac{1}{2}(\vec{AV} + \vec{UV}) = \vec{DA} - \frac{1}{2}(\vec{UV} + \vec{VC})$$

$$\cancel{\vec{DA}} + \frac{1}{2}\vec{AV} + \frac{1}{2}\vec{UV} = \cancel{\vec{DA}} - \frac{1}{2}\vec{UV} - \frac{1}{2}\vec{VC}$$

$$\vec{AV} = \vec{VC}$$

$$\vec{UV} = -\vec{AV}$$

quindi in modulo sono uguali e di conseguenza, se $|\vec{UV}| = |\vec{AV}|$ e $|\vec{AV}| = |\vec{VC}|$ ho demonstrato che $\overline{DH}$ e $\overline{KB}$ dividono il segmento in tre parti uguali.



$$A = l\hat{j}$$

$$D = l\hat{i} + l\hat{k}$$

$$\vec{AD} = \vec{AO} + \vec{OD} = (l, -l, l)$$

$$\vec{OA} = (0, l, 0)$$

$$\vec{OD} = (l, 0, l)$$

$$AD^2 = \vec{AD} \cdot \vec{AD}$$

$$AD = \sqrt{\vec{AD} \cdot \vec{AD}}$$

$\vec{AD} \cdot \vec{AD}$ è il prodotto riga per colonna

$$(l \ -l \ l) \begin{pmatrix} l \\ -l \\ l \end{pmatrix} = l^2 + l^2 + l^2 = 3l^2 \quad \text{allora} \quad \sqrt{3l^2} = l\sqrt{3}$$

